

Ayush Bansal

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SUMMARY

Ph.D. student at UIUC working at the intersection of **quantum systems** and classical HPC infrastructure. My research focuses on **Quantum-Centric Supercomputing (QCSC)** – co-designing the software stack that schedules quantum workloads across CPUs, GPUs, and QPUs. My current emphasis is on making **Quantum Error Correction (QEC)** pipelines efficient enough for **real-time** operation. I bring a strong systems background spanning **operating systems**, distributed systems, **low-latency FPGA**-based trading infrastructure, and **real-time systems**, which informs how I reason about the classical compute bottlenecks.

Languages : C/C++, Python, Golang, Scala, Rust, JavaScript/TypeScript, Haskell, Ruby, Java, Verilog

Web : Akka (Scala), Gin (Golang), Node.js, React, Flask, Ruby on Rails, Django

Utilities : Docker, Kubernetes, eBPF, Ansible, AWS, Azure, Kafka, GDB, SQL, MongoDB, $\text{\LaTeX}$, Xilinx ISE, Xilinx Vivado

Quantum : Qiskit (IBM Quantum), Stim, Munich Quantum Toolkit (MQT), Cirq (Google Quantum AI)

EDUCATION

University of Illinois Urbana-Champaign (UIUC)

Ph.D. in Computer Science (CGPA: 4.0/4.0) (advisor: Prof. Tianyin Xu)

2024 – Present

Urbana, Illinois

Indian Institute of Technology Kanpur (IITK)

B.Tech. in Computer Science and Engineering (CGPA: 9.0/10.0)

2016 – 2020

Kanpur, India

PROJECTS

Improving Relay-BP Performance using RL

May 2026 – Present

Dr. Apoorve Mohan and Dr. Shraddha Singh

IBM

- Relay-BP is state-of-the-art for qLDPC codes but is highly sensitive to configuration (γ) parameters; our goal is to train a RL agent to dynamically set (γ) parameters based on the syndrome data stream in real-time to achieve faster convergence and improve accuracy.
- Analysing syndrome datasets from Stim resembling real noise; observed inter-syndrome correlations suggesting learnable patterns.

Understanding Classical Resource requirements for QCSC pipelines

Feb 2026 – Present

Prof. Tianyin Xu and Dr. Apoorve Mohan

UIUC

- Built a QEM profiler exposing multiple optimization dimensions in the Quantum Error Mitigation pipeline; designing an end-to-end workflow manager for quantum workloads over a shared HPC stack (CPUs, GPUs, QPUs).

My way or the PHY way: Every Packet deserves the right MCS

Jan 2025 – July 2025

Prof. Brighten Godfrey and Prof. Matthew Caesar

UIUC

- Designed an application-aware rate-control mechanism optimizing per-packet MCS based on traffic intent (latency vs throughput).
- Cut tail latency for real-time traffic by eliminating mandatory probing (10–15% overhead) used in algorithms like Linux Minstrel, exposing application semantics to the PHY/MAC layers.

Olympus: Semantic query framework for satellite imagery

Jan 2025 – Dec 2025

Prof. Indranil Gupta

UIUC

- Engineered a real-time semantic query framework for satellite imagery, shifting image-centric retrieval to representation-centric.
- Integrated geospatial foundation models (SatlasNet) for onboard embeddings, achieving a 100x downlink reduction and enabling direct parallel querying without image access.

EXPERIENCE

International Business Machines (IBM)

May 2026 – Present

Research Extern

Yorktown Heights, NY

- Research on real-time Quantum Error Correction: improving QEC decoding by developing a framework to configure the relay-bp decoder in real-time based on incoming syndrome data stream from the quantum system.

Siebel School of Computing and Data Science, UIUC

Graduate Research Assistant

Aug 2024 – Present

Urbana, Illinois

- Research on Quantum-Centric Supercomputing: co-designing quantum-classical systems that schedule quantum workloads across CPUs, GPUs, and QPUs to maximize fidelity at least resource cost.

NimbleEdge

Co-Founder and Chief Technology Officer

Mar 2021 – Feb 2024

Bangalore, India

- Raised a \$3.3M pre-seed (Neotribe Ventures, Sistema Asia Capital) and built the entire technology org, migrating compute from the cloud to edge devices.
- Onboarded OYO, Dream11 & Glance (InMobi) as clients (each 100M+ MAUs); built a P2P SDK and aggregation infrastructure scaling to hundreds of millions of devices.

Tower Research Capital

Low Latency Software Developer

Aug 2020 – Mar 2021

Gurgaon, India

- Started as an Intern, earned a Pre-Placement Offer and continued as a full-time engineer.
- Owned the firm-wide Trade Reconciliation System across all markets, ensuring correct functioning of the trading platform and handling FPGA & network errors; extended it to Taiwan Futures and crypto exchanges (FTX, Huobi).
- Containerized and automated the Trade Reconciliation setup as microservices on bare-metal Kubernetes.
- Built low-latency C++ software inducing deterministic mutations on the network stream to simulate trading-platform and FPGA failure scenarios significantly reducing production incidents and potentially saving millions of dollars.

InterviewVector

Co-Founder and Chief Technology Officer

June 2020 – Aug 2020

Gurgaon, India

- Created Interview-as-a-Service category; signed 10+ clients (BharatPe, MindTickle) and bootstrapped to \$170K ARR in 4 months.

Eightfold.ai

Software Developer Intern

Apr 2020 – June 2020

Noida, India

- Built the Alert Management System & set up BDD testing with Selenium, cutting test-suite runtime from 62 to <10 minutes (84%).

Google Summer of Code — OpenPrinting, The Linux Foundation

Open Source Developer

May 2018 – Aug 2018

Remote

- Developed a Common Print Dialog backend serving all available print queues (CUPS, Google Cloud Print) over D-Bus.

PUBLICATIONS

Rex: Closing the language-verifier gap with safe and usable kernel extensions. Jinghao Jia, Ruowen Qin, Milo Craun, Egor Lukyanov, **Ayush Bansal**, Michael V. Le, Hubertus Franke, Hani Jamjoom, Tianyin Xu, Dan Williams. *USENIX Annual Technical Conference (ATC '25)*.

ACADEMIC SERVICE

Artifact Evaluation Committee

Aug 2024 – Present

- OSDI '25 · USENIX ATC '25 · SOSP '25 · ACM SIGCOMM '25 · EuroSys '26.

HONORS & AWARDS

NSDI Student Grant Award — travel grant to attend the conference	2025
Academic Excellence Award — exceptional academic performance, 3 consecutive years (IITK)	2020
Exceptional Course Tutor — rated exceptionally by students, AY 2019–20 (IITK)	2019
JEE Advanced — All India Rank 187 out of 200,000 candidates	2016
KVPY Scholarship — All India Rank 52 (IISc & Government of India)	2016
JEE Mains — All India Rank 571 out of 1.5 million candidates	2016
Indian National Chemistry Olympiad — Merit Certificate, National Top 1%	2015
Indian National Astronomy Olympiad — Merit Certificate, National Top 1%	2015